

Final Project

113370450 蘇孟孜

1. Research Question

Does sleep duration predict the likelihood of feeling sad or hopeless among high school students?

2. Dataset & Variables

Dataset: CDC 2007 YRBS (N = 12,106 after data cleaning)

Independent Variable

Sleep Duration (Type: Ordinal)

1 = ≤ 4 hours

2 = 5 hours

3 = 6 hours

4 = 7 hours

5 = 8 hours

6 = 9 hours

7 = ≥ 10 hours

Dependent Variable

Sad_Recoded

1 = Yes

0 = No

(Type: Binary)

3. Method

Logistic Regression

Sleep $\rightarrow$ Sad_Recoded

Hypotheses

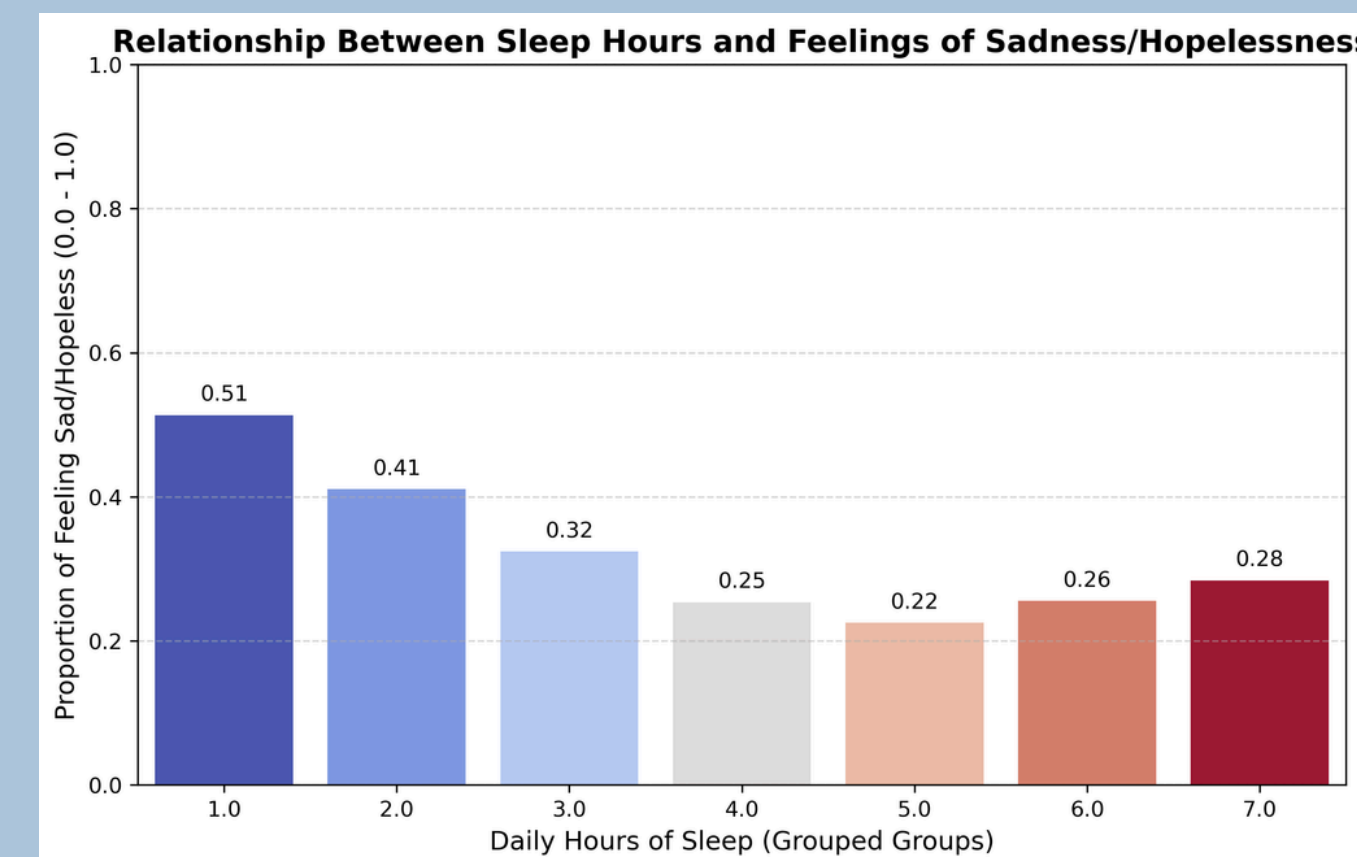
H₀: Sleep duration does not significantly predict sadness or hopelessness.

H₁: Sleep duration significantly predicts sadness or hopelessness.

$\alpha = 0.05$

4. Key Result

Statistic	Result
N	12,106
β	-0.2475
OR	0.7807
p-value	<0.001



5. Conclusion

Sleep duration was a significant predictor of feelings of sadness or hopelessness among high school students ($p < 0.001$). Longer sleep duration was associated with lower odds of reporting sadness or hopelessness.